

116_aNCA_workflow_summary.Rmd

Jim

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News: This fellow's work looks superior to my own attempt. Need to review! <https://teunp.github.io/PKNCA.doc/user/workflow.html>

[TODO:

- intervals - better discussion, examples.
- explain "formula" (conc-time | group) and how it relates to grouping keys, partitioning, profiles, etc.
- annotate the Theoph example
- be certain "PKNCA" dataset is clear.
- SDTM as input: <https://github.com/humanpred/pknca/issues/516> (I like)
- issue #289 and this document: <https://www.lexjansen.com/pharmasug/2017/PO/PharmaSUG-2017-PO01.pdf> [.blue]

[To reader: This is intermediate work-in-progress and **not ready for serious review**

This vignette began as my own raw notes. Although the idea behind aNCA seemed very clear, number of variable names, CDISC's role, and general data flow constantly confused me. [.blue]

The goal here is to: - depict the data flow, according to CDISC ideas. - then understand how PKNCA functions fit into the this data flow - at deeper level, understand how the PKNCA transform data at each step, and how the final output tables are generated - in later step, depict the data flow beginning with aNCA and shiny (not yet done)

Intro

These are my notes. As such very imperfect with errors, gaps and missing logic.

The original intent was to bridge the gap between 2 vignettes: 1. background.md 2. this document 3. manual.md

This document is to show high-level, CDISC aligned work-flow from raw data to final ADaM, meaning ready for TLG.

The second section goes a little deeper ideas in the **implementation**. Vignette 3 (manual.md) does excellent job working through the categories and details of dataset variables. This section is intended to introduce the workflow ideas. A final section goes a little deeper into functions, tibbles.

The goal is to ease the transition to code-level and the manual.md

Level 1: Concepts, CDISC

The Data Pathway

TODO: re-write with CDISC The lifecycle of PK data follows a rigid, standardized progression:

The clinical reporting pipeline is an ETL (Extract, Transform, Load) architecture divided strictly into Pre-SDTM and Post-SDTM phases.

pre-SDTM

Raw Concentration Stream → [Custom R ETL Scripts] → SDTM PC Table & SDTM EX Table Raw Dosing Stream /

Pre-SDTM Mapping (Normalization & Standardization) TODO: Clean up!

Data arrives as raw, unstructured database tables or flat files from separate transactional systems. To get these to a valid CDISC SDTM format, a clinical programmer must build custom R mapping scripts.

The Dosing Table Pipeline:

Raw fields from the clinic’s Electronic Data Capture (EDC) system are mapped to the SDTM EX (Exposure) domain.

The Concentration Table Pipeline:

Raw transactional fields from the external Bioanalytical Lab are mapped to the SDTM PC (Pharmacokinetic Concentrations) domain.

Primary & Composite Keys in SDTM PC

Because clinical data is highly denormalized, no single field guarantees row uniqueness. You must look at the data through composite keys.

The Business/Natural Composite Key:

To uniquely identify a concentration record under the official CDISC standard, you require a composite key of:

{STUDYID, USUBJID, PCTESTCD, PCCAT, PCVISITNUM, PCDTC, PCTPTNUM}

The Foreign Key (PCSEQ): Real-world operational data can capture duplicate timestamps (PCDTC). To guarantee absolute entity integrity, CDISC enforces a dedicated foreign counter field, PCSEQ. Therefore, the leanest guaranteed technical key for any row in the table is:

{USUBJID + PCSEQ}

Raw Data → SDTM (PC/EX) → aNCA Engine → ADaM (ADPC/ADPP)

- **Raw Data:** Consists of one and possibly two separate, messy streams: **Dosing** (from clinical sites) is required and **Concentrations** is optional (from bioanalytical labs).
- **SDTM:** The “Rosetta Stone” phase where a programmer maps those two streams into a uniform language using strict, standardized variables (e.g., USUBJID, PCSEQ).
- **ADaM:** The final “analysis-ready” structures (**ADPC** for concentrations/curves, **ADPP** for calculated macro-parameters like C_{max} and AUC).

-
- **Who Maps Raw Data to SDTM?**
 - **aNCA** does **not** touch raw data. The mapping to SDTM is done beforehand by a **Programmer** (user? , other pharmaverse packages?) using standard R/tidyverse tools, guided by a study specification document.

Table A: SDTM PC (Pharmacokinetic Concentrations)

This table acts as a time-series log tracking drug levels found in blood samples.

- **Natural Composite Key:** STUDYID + USUBJID + PCTESTCD + PCCAT + PCVISITNUM + PCDTC + PCTPTNUM
- **Technical Surrogate Key:** USUBJID + PCSEQ (Where PCSEQ is a sequential auto-incrementing integer starting at 1 for each unique subject).

# Conceptual Example of a Clean SDTM PC Dataset								
	STUDYID	USUBJID	PCTESTCD	PCORRES	PCORRESU	PCVISITNUM	PCDTC	PCSEQ
1	PROT-001	SUBJ-101	DRUG-X	BQL	ng/mL	1	2026-10-12T08:15	1
2	PROT-001	SUBJ-101	DRUG-X	10.5	ng/mL	1	2026-10-12T09:15	2
3	PROT-001	SUBJ-101	DRUG-X	45.2	ng/mL	1	2026-10-12T10:15	3
4	PROT-001	SUBJ-101	DRUG-X	22.1	ng/mL	1	2026-10-12T12:15	4
5	PROT-001	SUBJ-101	DRUG-X	5.3	ng/mL	1	2026-10-12T16:15	5

(Note: `BQL` stands for Below the Quantifiable Limit-it means the lab instrument detected the drug but it was too low to measure as a c

aNCA expects to find its source data stored in two standardized tables defined by the CDISC SDTM framework. Because a clinical database is wide and flat rather than fully normalized, unique rows are enforced using composite natural keys or a dedicated technical surrogate key.

Table B: SDTM EX (Exposure / Dosing) - *Optional*

This table tracks exactly when a physical dose of the drug was administered to the patient.

- **Technical Surrogate Key:** USUBJID + EXSEQ

# Conceptual Example of an SDTM EX Dataset							
	STUDYID	USUBJID	EXTRT	EXDOSE	EXDOSU	EXSTDTC	EXSEQ
1	PROT-001	SUBJ-101	DRUG-X	100	mg	2026-10-12T08:15	1
2	PROT-001	SUBJ-102	DRUG-X	100	mg	2026-10-12T08:30	1

post-SDTM -> ADaM

Do not use aNCA, say “processing” PKNCA is doing the processing

Post-SDTM Phase:

The aNCA application begins execution *strictly* after this boundary. It reads the standardized input tables, blends them with an external configuration, and generates completely new relational analysis tables. It cannot touch the raw data.

SDTM PC & EX Tables —> [aNCA Engine + Configuration Metadata] —> ADaM ADPC & ADPP Tables

- **What aNCA Expects as Input:**
- aNCA expects the data to **already be in clean SDTM format** before it ever reads it. aNCA requires SDTM PC (concentrations) and optionally accepts SDTM EX (dosing).
- **The “Recipe” (Analysis Settings):**
- The R code automates the *execution*, but the **Programmer** must supply the *rules*, for example *exclusions* (more later). You must pass aNCA functions your specific **Analysis Settings** (arguments/metadata) to tell it exactly how to handle BQL values, how to group profiles, and which PK parameters to calculate. (ex: Cmax)

(TODO: mapping SDTM -> ADaM colNames)

Source SDTMColumn	Target ADaMColumn	TransformationType	ConceptualPurpose
PCORRES (Char)	AVAL (Numeric)	Type Cast / Imputation	Converts text strings to float values for math engines.
PCTESTCD (Char)	PARAMCD (Char)	System-wide Remap	Converts a lab test code to an analysis parameter code.
PCDTC / EXSTDTC	AFRLT (Numeric)	Calculated Delta	Computes Actual Relative Time from Analyte First Dose (tsample–tdose).

It imports the valid SDTM tables and handles a complex mapping and derivation layer to generate CDISC ADaM (Analysis Data Model) datasets. This step requires a complete schema migration where source variables are transformed, re-typed, and renamed into analysis-ready targets: Source SDTM ColumnTarget ADaM ColumnTransformation TypeConceptual Purpose

Once aNCA package receives the SDTM data and your Analysis Settings, it alters the dataset by renaming variables (e.g., PCORRES → AVAL), imputing BQLs, and adding relative time columns. Its very first task is **Profile Grouping**:

- It isolates unique time-concentration curves.
- *Crucial Concept:* If the same patient returns for a second visit, aNCA treats this as a completely separate, independent PK profile (Curve 1 vs. Curve 2).

The Rules Engine: Configuration Metadata

Redo this entire section!

The aNCA R code executes transformations programmatically, but it lacks the logic to make study-specific data decisions on its own. The developer must supply these rules via a **Configuration Metadata Object** (instantiated as an R list object or a JSON configuration file read into the R session memory at runtime).

This metadata does not mix into the source tables; it acts as a look-up parameter matrix that dictates processing logic:

- **Data Partitioning Blueprint:** Instructs the R engine which columns to group by to build an independent time-concentration matrix (e.g., partition data by USUBJID + PCVISITNUM + PCTESTCD).
- **Imputation Logic Rules:** Contains explicit conditional mapping instructions for the math engine. For example: “If a row contains PCORRES == 'BQL' AND the sample timestamp PCDTC is less than or equal to the dose timestamp EXSTDTC, mutate the numerical result to 0. If it occurs after a valid numeric concentration, mutate it to NA.”

Instead of being hardcoded, this is treated as a Configuration Metadata Object (or a settings file uploaded into the aNCA session). Where it sits in the Data Flow

The metadata object acts as a side-car lookup parameter. It does not replace the source data; it is read simultaneously by aNCA processing functions to determine conditional branching logic during data processing.

SDTM PC Table ———> [aNCA Engine] ———> ADaM ADPC/ADPP Tables ^ Configuration Metadata —+ (Controls logic for BQL, Grouping keys, Slope rules)

What this Metadata contains:

Imputation Rules (e.g., BLQ/BQL): Defines conditional value replacements. For example, if a record is flagged as Below the Limit of Quantification

Partitioning/Grouping Keys: Tells the R script which column combinations constitute a distinct, isolated time-concentration curve (a single

Output Tables (ADaM Target Schema)

By executing the `aNCA` processing functions, the R engine mutates and renames fields to conform to CDISC ADaM standards. The two primary target tables generated are:

Table 1: ADPC (Analysis Dataset PK Concentrations)

This is an altered, enhanced clone of the input concentration data. Text fields are cast to numeric floats, non-numeric strings are imputed, and absolute date-timestamps are calculated into relative numeric intervals.

- `PCORRES` (**Character**) is converted and renamed to `AVAL` (**Numeric**).
- `PCTESTCD` (**Character**) is converted and renamed to `PARAMCD` (**Character**).
- The delta between `PCDTC` and `EXSTDTC` is calculated to create a new numeric time elapsed variable, `AFRLT` (Actual Relative Time).

#	Conceptual	Example of a Generated ADPC Table				
	USUBJID	PARAMCD	AVAL	AVALU	AFRLT	VISITNUM
1	SUBJ-101	DRUG-X	0.0	ng/mL	0.0	1 <-- Imputed BQL to 0.0
2	SUBJ-101	DRUG-X	10.5	ng/mL	1.0	1 <-- Calculated 1 hour post-dose
3	SUBJ-101	DRUG-X	45.2	ng/mL	2.0	1
4	SUBJ-101	DRUG-X	22.1	ng/mL	4.0	1
5	SUBJ-101	DRUG-X	5.3	ng/mL	8.0	1

Table 2: ADPP / ADNCA (Analysis Dataset PK Parameters)

This table stores the final summarized results calculated across an entire partitioned data group. Instead of representing individual blood draws over a timeline, it aggregates a single profile partition down to **one unique summary row per parameter, per patient profile**.

- Technical Surrogate Key:** `USUBJID` + `PARAMCD` + `PPSEQ`

#	Conceptual	Example of a Generated ADPP Summary Table				
	USUBJID	PARAMCD	PARAM	AVAL	AVALU	VISITNUM
1	SUBJ-101	CMAx	Max Concentration	45.2	ng/mL	1
2	SUBJ-101	TMAx	Time of Max Conc	2.0	hr	1
3	SUBJ-101	AUCALL	Area Under the Curve	185.4	h*ng/mL	1

Official Specification Blueprints

The core formatting rules, naming restrictions, and validation schemas used to govern these structural designs are fully documented in the following global clinical data registry standards:

- Input Tables Validation Schema:** [CDISC SDTM Implementation Guide \(SDTMIG\)](#)
- Output Tables Validation Schema:** [CDISC ADaM Implementation Guide \(ADAMIG\)](#)

““

For the exact data structures, column schemas, and validation constraints governing these transforms, the primary regulatory blueprints are:

Pre-SDTM Mapping Rules: Formally documented in the Data Transfer Agreement (DTA) or the study-specific Data Specifications Document (Data Sp

Official CDISC Standards:

To verify the structural rules of the input PC table, refer to the CDISC SDTM Implementation Guide (SDTMIG).

To verify the structural rules of the output tables, refer to the CDISC ADaM Implementation Guide (ADAMIG).

Level2: PKNCA implementation

Overview

To implement the ideas and pathways of CDISC (SDTM, ADaM), the package PKNCA does much of work to calculate NCA. Its implementation takes advantage of many R features to simplify coding (example: R objects) Even when PKNCA gets to the place CDISC expects, it is not always obvious how align both paths.

In this section, outline of the PKNCA’s approach. In final section, get to code.

(TODO) This paper https://phuse.global/Communications/PHUSE_Blog/r-handling-of-noncompartmental-analysis-pharmacokinetic-packages has succinct data flow and function definition This paper “<https://deepwiki.com/humanpred/pknca/1-overview>” has data flow.

(TODO: introduce core functions, the objects and data flow in PKNCA implementation.)

PKNCA functions, objects, flow

[TODO: specifics]

